

3DM 2024: Theorems whose proofs might be on the exam

This is a list of theorems whose proofs are in the notes that you must be able to prove on the exam. Note that you must know the statement of every theorem we covered in class, whether or not the proof is examinable.

1. Theorem 1 (First Theorem of Graph Theory)
2. Corollary 2 (Every graph has an even number of odd vertices)
3. Theorem 3 (Havel-Hakimi)
4. Theorem 5 (Every $u - v$ walk contains a $u - v$ path)
5. Theorem 7 ($\text{rad } G \leq \text{diam } G \leq 2 \text{ rad } G$)
6. Theorem 8 (Bipartite graphs and odd cycles)
7. Theorem 11 (Bridges and cycles)
8. Theorem 13 (Blocks and cycles)
9. Theorem 15 (The center lies in a single block)
10. Theorem 16 (Every nontrivial tree has ≥ 2 endvertices)
11. Theorem 18 (For a tree, $m = n - 1$)
12. Theorem 21 (Equivalent statements for a tree)
13. Theorem 25 (Euler's Formula)
14. Theorem 27 (For a maximal planar graph, $m = 3n - 6$)
15. Lemma 29 (Planar graphs have $\delta(G) \leq 5$)
16. Theorem 33 (Chvátal's Art Gallery Theorem)
17. Theorem 35 (Fáry's Theorem)